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A SERM based framework to optimize the modelling of entity life history

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Abstract

This article is the next in a series presenting the application of the SERM (Structured Entity Relationship Model), an extension of the classic ERM (Entity Relationship Model). An in-depth analysis of the model reveals the enormous possibilities offered by its hierarchical layout. So far, the SERM model has only been used at the data modelling stage. This technique allows the construction of semantic data structures, the identification of mandatory relationships, data blocks and logical rules to secure data integrity.

Based on the assumption that an information system is used to process data, all the events that support it and their relationships with children and/or parents form the story of their lives. By analogy with human life, the starting point for creating entity life history diagrams is the basic sequence of events: birth, growth and death.

The aim of this paper is to present new functionalities of a framework based on the SERM model. The proposed solution supports the intelligent modelling of events that form the life history of entities. In this way, it goes beyond the scope of data modelling and eliminates the need to build a model of functions and processes.

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1. Introduction

Analysis is the first phase of information systems development. Its primary task is to understand the problem domain (interpretation of the requirements of potential users of the system) and to determine the responsibility of the

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planned system against this background. In effect, the analysis leads to an answer to the question "*what should we computerize?*" rather than "*how should we computerize?*" [19].

In software engineering, there are two approaches to system analysis and design:

- object-oriented (OOA&D)[3],
- structural (SA&D).

Both concepts have different methods and notations resulting from different approaches to data representation and system functions [11]. Object-oriented approach characterized by common representation data and functions (processes) by introducing the concept of object and class. According to the definition of the creators of this approach, P. Coad and E. Yourdon, the object is a "capsule with attribute values and services operating on them" [6,7]. In the structured approach, data and functions are treated as separate problems. Data modeling and modeling of functions and processes are carried out independently of each other, often by different project teams. In addition to the data model and functional model, the structural methodology describes the system in the third aspect. This is served by dynamic models, which represent events and/or states of the designed system based on previous realizations of the data model and the function and process model [19]. One of these is the entity life history (ELH) model, which is the subject of the presented research.

The aim of this paper is to present a method for optimizing event modelling based on the SERM (Structured Entity Relationship Model). The existing approach to entity life history in the form of ELH diagrams requires the implementation of a function and process model [2]. This is due to the need to build an object-event cross-tabulation as described in Chapter 2. The novelty of the proposed author's method is the absence of this requirement. The main focus is on the semantic data model. This is an essential element for further analysis. A faithful representation of reality will allow the generation of functions to handle the modelled data, since the main functionality of any information system being designed is data processing.

The SERM model was first presented in 1989 by prof. Elmar J. Sinz from the University of Bamberg (Germany) [23, 24]. It is an extension of the classic ER model [5]. What particularly distinguishes it and is the object of the author's interest is the hierarchical arrangement of the diagram. Each object in the diagram has its own specific place in the hierarchy as a result of relationship analysis. SERM diagram analysis formed the basis for the solutions described below. This article presents a framework to support the creation of object life histories based on the SERM model. The development of a new method is the result of continuing work on SERM diagram analysis. Undoubtedly, the author exploits the research gap in this topic of the possibilities it offers. Descriptions of diagram analysis in general are lacking in the literature.

The article is divided into five parts. After an introduction, Section 2 presents the specification of object life history modelling. Section 3 presents the block diagram of the framework together with detailed algorithms for the operation of the individual blocks. A presentation of the practical application of the developed framework is the content of Chapter 4. It provides a case study on the example of a Patient Service System. The last part of the article is a summary with conclusions indicating possible further directions of work.

2. Modelling the history of events

Basic a technique for modeling the history of events in the structural approach there are diagrams ELH (Entity Life History). Modeling of event history using the ELH technique requires previous data model implementations (ERD/SERM) and the function and process model (DFD). Diagrams ELH show a third plane of vision IT system. It is event-based (defined in the DFD diagram) that interact to objects defined in the ERD or SERM diagram [11, 25]. Event history modeling using the ELH technique is being implemented in two stages:

- construction entity-event crossover table,
- construction ELH diagram for each object.

The entity-event crossover table is created as a result of assigning events from the DFD model to each object from the ERD/SERM model. So it shows all the objects and all events affecting them. Table 1 shows the form of a

table for a system with N entities defined and M events identified. For example, in the life history of Entity 1, three events should be included: Event 1, Event 2 and Event M .

Table 1. Form of an entity-event crossover table

Entity\Event	Event 1	Event 2	Event 3	...	Event M
Entity 1	X	X			X
Entity 2					
Entity 3			X		
....	X	X			
Entity N		X		X	

Based on the table, an ELH diagram is created for each entity. This should be done in three steps:

- modeling the normal life cycle of an object,
- supplementing it with unique (unusual) events,
- taking into account the occurrence of errors and emergency events.

There are two types of events in the ELH diagrams:

- simple types,
- complex types.

Simple events are divided into:

- ordinary - a single event executed once,
- iterative - a single event executed multiple times.



Fig. 1. Designations of simple event types.

On the other hand, among complex events we distinguish:

- sequential - a branch of events where the condition for the execution of the next event is the execution of the previous one (executed from left to right),
- selective - a branch of events, among which only one event will be executed.

A special case of an event is an empty event, used in the selective events branch.

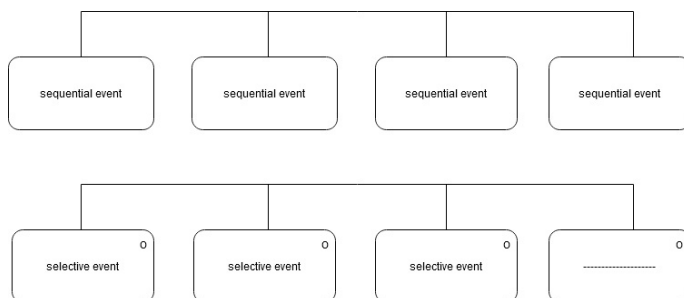


Fig. 2. Designations of complex event types.

Single the ELH diagram focuses on one thing selected object (object type) defined in using data modeling techniques (e.g. SERM) and presents the "fate" of its mortgage instance. History the life of an entity, by analogy to human life, is presented as basic sequence:

- birth - creating/adding entity,
- growth - entity operations,
- death - entity deactivation.

3. Framework to optimize the events identification

The hierarchical arrangement of objects typical of SERM diagrams and the preservation of referential integrity have a significant impact on the order in which data is entered into objects and processed in the system. This has been exploited in the optimization of the identification of obligatory relations, which has been one of the main functional elements of the presented framework so far.

An in-depth analysis of the topic made it possible to assume that the "fate" of an object is strongly dependent on its position in the diagram. Furthermore, its life history is strongly influenced by its membership in a data block. Resource blocks and complementary blocks are two of the three pillars on which the method of optimizing the occurrences of obligatory relations is based [14, 16]. In the case of a resource block, it matters whether an entity is a key element or not. This has implications for the modelling of its child life history. When we create an object and its parent is a resource, we only consider one event: the selection of a parent from a list. This is because the resource is previously entered into the database and updated independently [1, 9]. There is therefore no option to add a parent.

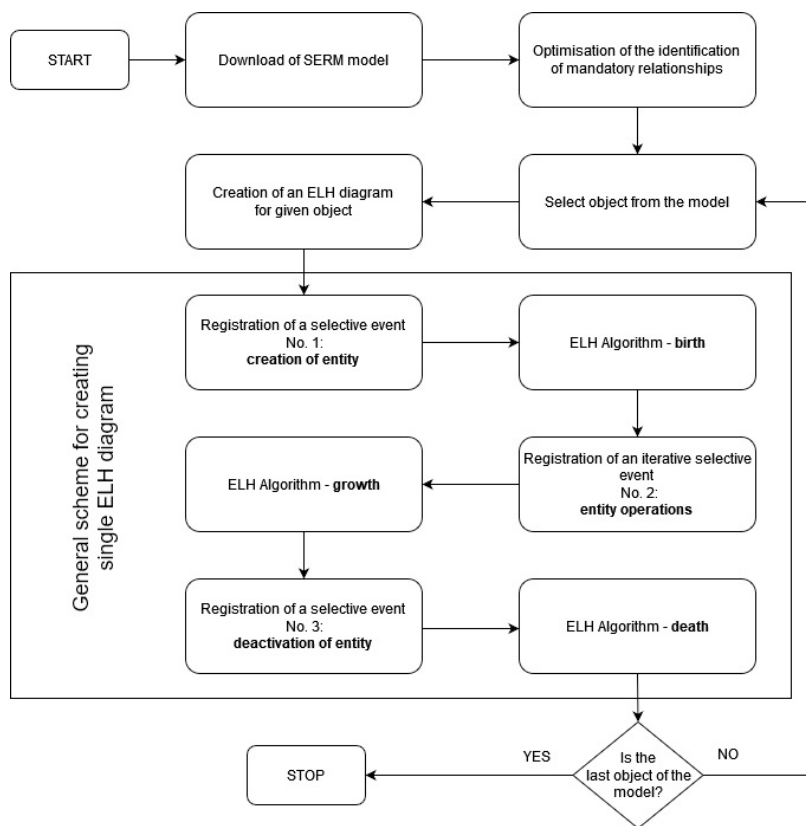


Fig. 3. General scheme for creating ELH diagrams.

In the complementary block, on the other hand, it matters whether the analyzed object has the status of an incomplete parent, as this implies an mandatory relationship. Forcing the occurrence of a child relationship has a significant impact on the object's event history. Such a relation must already be taken into account at the object creation level. A single entity is not entered into the database directly by means of an INSERT statement, but by means of a procedure within a single transaction.

The presented method for optimizing event identification can be an extension of the functionality of the SERM-based framework. Under this assumption, the identification of data blocks (resource and complementary) will no longer be necessary. Otherwise, the activities of the event identification optimization method must be preceded by the application of the method to optimize the identification of the mandatory relationship, as can be observed in the algorithm shown in Figure 3.

An ELH diagram is created for each object from the analyzed SERM model. The general chart for creating a single ELH diagram, extracted in Figure 3, is based on a template reflecting the main life sequence (Figure 4).

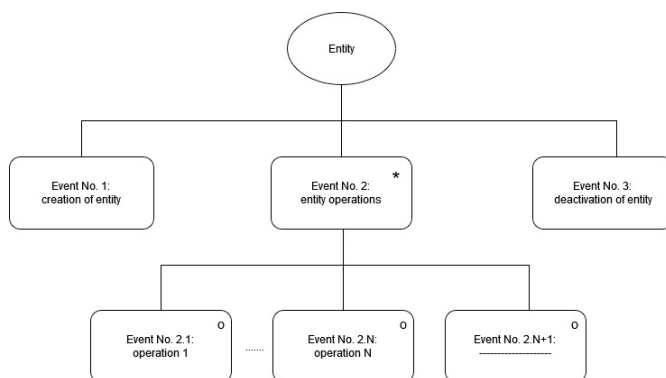


Fig. 3. ELH diagram template.

Once each of the three main sequence events has been registered, the template event expansion algorithms are implemented. The algorithm that generates the entity creation events is shown in Figure 5. Depending on the degree of inheritance hierarchy (level higher than 1) and the role of the parent (with a mandatory relation), a sequence of events is created that trigger the creation of a new object. In the case of the child object of the key parent, it is sufficient to register the event of selecting the parent from the list. When the parent is not a resource, the possibility of creating a parent object as a selective event with the selection of a parent from a list should also be included. In addition, a “introduction a relationship with the child” event should be created for each mandatory relationship. This will force the design of a procedure (handled by a single transaction) that creates an object simultaneously with its child.

According to the template, the middle event type (entity operations) will always be an iterative event that will develop into a branch of selective events, i.e. executable object operations and an empty event. The scheme of operations is shown in Figure 6.

The number of selective events generated depends on:

- editability of own attributes,
- editability and number of foreign keys,
- number of child relationships.

Editing of custom attributes is usually modelled in a single event regardless of the number. In contrast, a separate event is created for each editable foreign key. As in the case of object creation, we take into account whether the parent is a key element or not, because any additional event selection depends on this. And when it comes to child relationships, a separate event must be modelled for each.

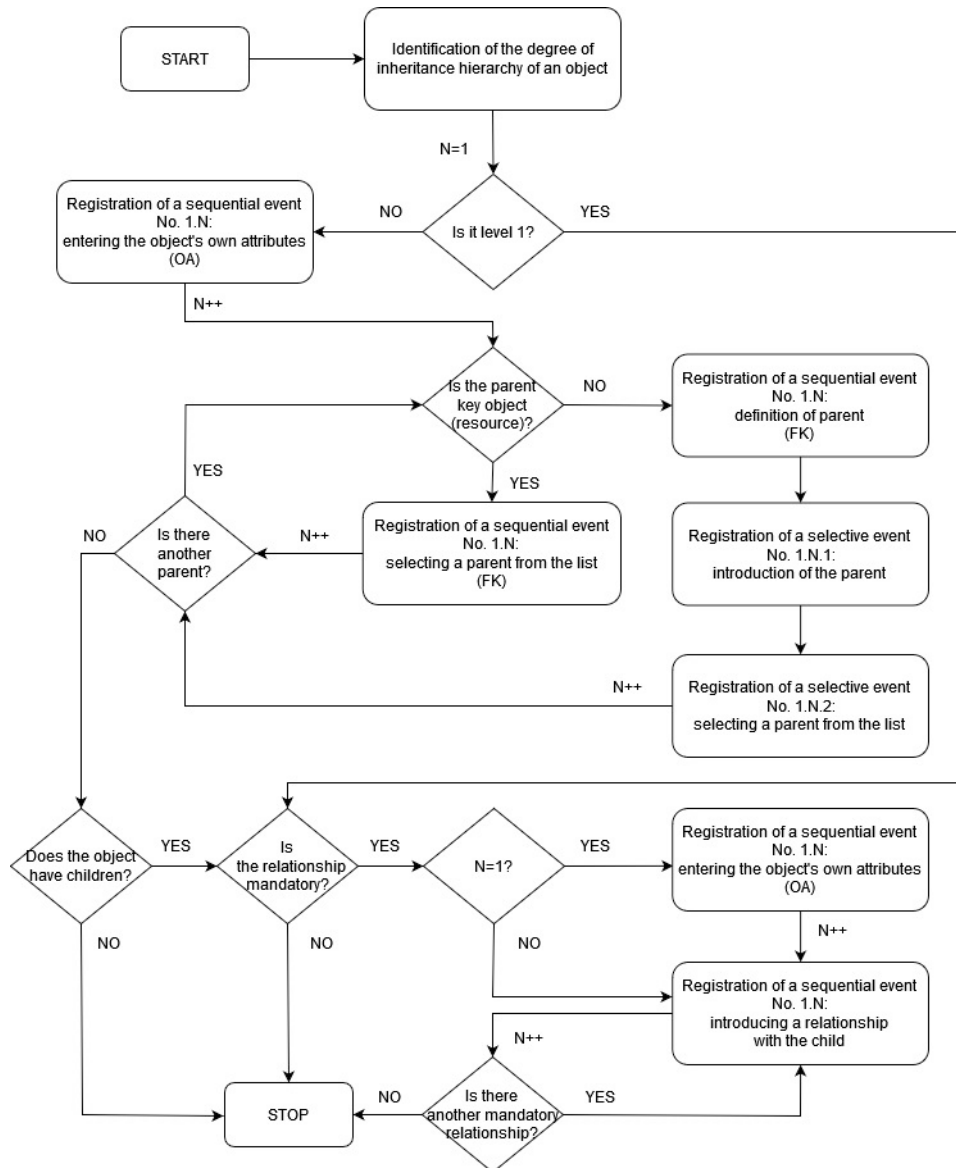


Fig. 5. ELH Algorithm - birth.

The third event “deactivation of entity” can become a simple event or develop into a branch of selective events, as shown in the algorithm in Figure 7.

If there is no need to develop the event i.e. there is no option to end the life of the entity, event number 3 can be renamed to one of the following:

- deleting the object,
- setting the activity flag to NO,
- archiving the object.

Otherwise, create a selection of two or all three of the above events.

4. Case study - model of the Patient Service System

As with previous methods based on the SERM model, the SERM diagram of the Patient Service System was used as input to demonstrate the practical use of the framework [16, 17]. In this case, the relationship complexity labels in (min;max) [14] notation play an important role, as shown in Figure 8. Due to the large number of objects of the modelled system (32 entities, entityrelations and relationships), it is not possible to present the generated ELH diagram for each of them in this paper. The operation of the method will be demonstrated on a selected, most representative object.

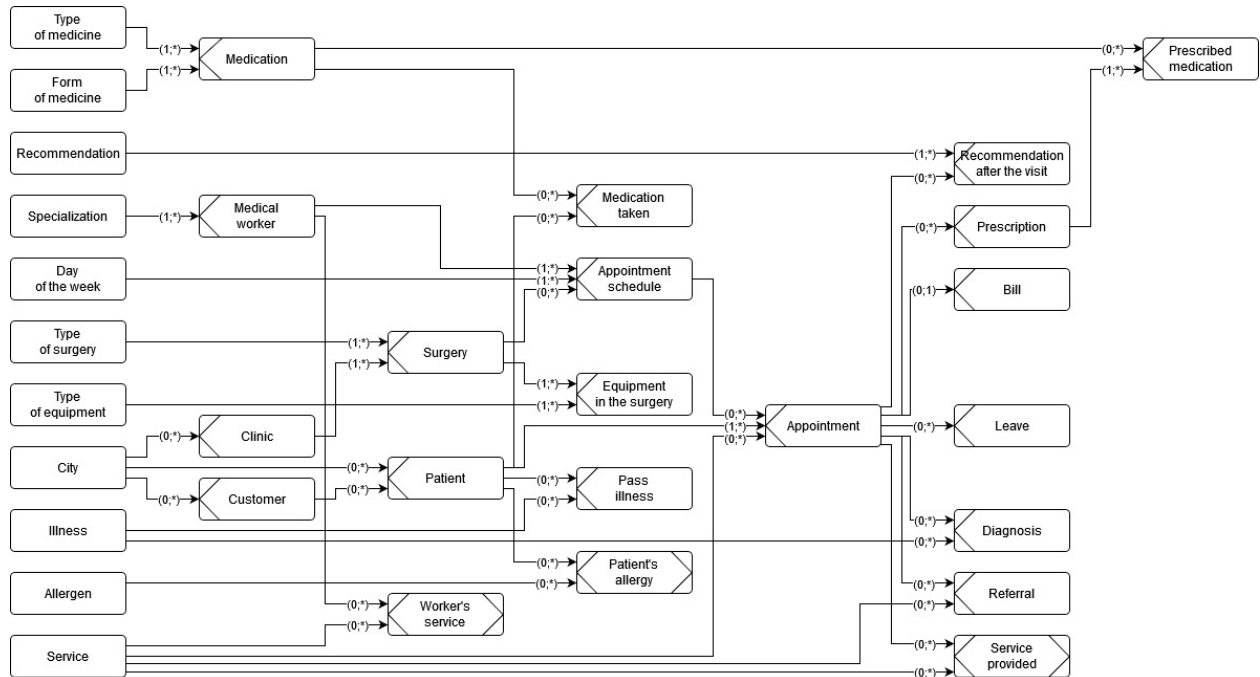


Fig. 8. SERM diagram for Patient Service System [16].

In the diagram, the object “Appointment” stands out. Due to the number of parents and children and its strategic importance, its ELH diagram will undoubtedly be a good example of the application of the method presented in the article.

The object being analyzed is not at level 1, so we need to record sequential event 1.1: *entering the date and time (OA)*. We then check the parents sequentially for resource creation. In this case, we have three objects:

- Appointment schedule - a key element,
- Patient – not a key element,
- Service - a key element.

The first parent creates a resource so a sequential event 1.2: *selecting a schedule from a list* will be generated. Then, for the second parent, which is not a key element, we create sequential event 1.3: *patient definition*, selective event 1.3.1: *entering patient data* and selective event 1.3.2: *selecting patient from list*. A service is a key element and therefore, in accordance with the algorithm, sequential event 1.4: *selecting a service from the list* will be generated.

The next step is to control the relationship with the children, of which there are seven objects. We check whether there are obligatory relations. Here they are all optional so we do not create any further events.

The operation of the entity creation algorithm resulted in a sequence of four events, one of which developed into a selection of two events (Figure 9).

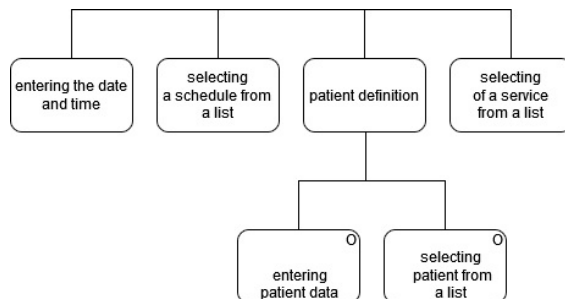


Fig. 9. Event of appointment creation.

The algorithm for entity operations starts by checking the editability of the object's own attributes and then the editability of its individual foreign keys. In the object under analysis, only the date and the service for which the patient is registering can be changed. Therefore, two events will be generated:

- selective event 2.1: *edition of date and time*,
- selective event 2.2: *edition of service*.

The next step is to record the relationship with each child, which in this case will give seven events:

- selective event 2.3: *entering a recommendation*,
- selective event 2.4: *writing a prescription*,
- selective event 2.5: *issuing of a bill*,
- selective event 2.6: *issuing of a leave*,
- selective event 2.7: *making a diagnosis*,
- selective event 2.8: *issuing of a referral*,
- selective event 2.9: *registration of ancillary service*.

At the last stage, according to the algorithm, selective event 2.10: *view/print data* and selective event 2.11: *empty event* should be generated.

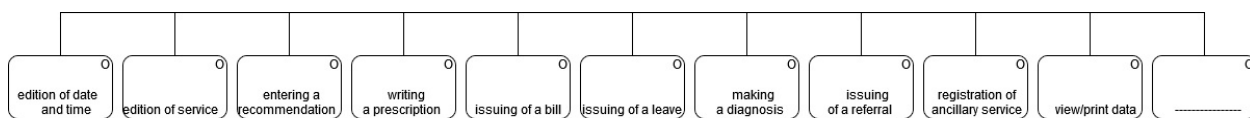


Fig. 10. Event of operations.

The task of the third algorithm is to generate events to handle the deactivation of entity. In the case of the “appointment” object, it should not be deleted, but the activity flag can be set to NO or the data archived. Therefore, two events will be created: selective event 3.1: *setting the activity flag to NO* and selective event 3.2: *archiving the visit*.

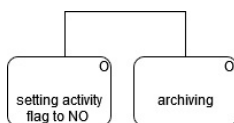


Fig. 9. Event of deactivation.

5. Summary

The original solution presented in the article aims to optimize the creation of dynamic models of the life history of an object. It extends the functionality of the developed framework based on the analysis of the SERM diagram. Existing solutions in the structural approach to analysis and design are characterized by high uncertainty and require the implementation of a model of functions and processes. The presented method is based on the assumption that the functionality of the system must result from data processing, because this is the main goal of creating IT systems. Therefore, it does not require a model of functions and processes, but is based on a semantic data model using the hierarchical arrangement of the SERM diagram. It assumes that the fate of an object is strongly dependent on its position on the diagram and its membership in data blocks (resource and supplementary), which is taken into account in the algorithms developed to handle individual stages of the life history of objects.

The solution presented here represents the third level of approach to this topic. The object-oriented paradigm has a common representation of data and processes in the form of classes and objects. In the structural approach, these are models treated separately, and the presented method generates events and system functionality based on the data model. Learning about the new possibilities offered by in-depth analysis of SERM models allows us to hope for new directions of research supporting the analysis and design of information systems.

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